Talha Zafar

Senior Backend Engineer | DevOps & Infrastructure Engineer

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Lahore, Pakistan | Open to on-site, hybrid, or remote

PROFESSIONAL SUMMARY

Backend and infrastructure engineer with **4+ years of production experience** designing and operating high-throughput distributed systems. Proven track record delivering microservices architectures, RESTful and gRPC APIs, and event-driven pipelines using **Python (FastAPI, Flask, Django)** and **Node.js (NestJS)**. Hands-on expertise in containerisation (Docker, Kubernetes/K3s), PostgreSQL/MongoDB data modelling, Redis caching, and cloud infrastructure on **AWS and GCP**. Deep DevSecOps background: bare-metal GPU cluster management, HashiCorp Vault secrets management, OWASP hardening, SAST/DAST pipelines. Recent work spans AI agent and LLM deployment platforms on bare-metal GPU clusters. Fluent in English (C1) and Urdu (native).

PROFESSIONAL EXPERIENCE

DevOps & Infrastructure Engineer

Jan 2026 – 2026

Augai · Infrastructure & Security Consulting · Remote

- Sole infrastructure owner for bare-metal Ubuntu 22.04 production server running a **4× NVIDIA Tesla V100 GPU cluster**, K3s Kubernetes, and 8+ LXD developer containers — maintaining **99.8% uptime**.
- Designed K3s cluster from scratch: nginx ingress controller, cert-manager with Let's Encrypt TLS automation, CoreDNS internal service discovery, and GPU passthrough configuration.
- Deployed **HashiCorp Vault** (AppRole auth, policy-based access control) as isolated LXD container for encrypted secrets management across CI/CD pipelines — **GDPR data residency compliant**.
- Executed full security hardening: forensic log audit (server confirmed clean), iptables stateful firewall, fail2ban brute-force protection, kernel update lockdown, rsyslog centralised logging across all containers.
- Built GitHub Actions CI/CD pipelines with integrated SAST (SonarQube), container image scanning (Trivy), and secret detection (Gitleaks).
- Configured Zabbix monitoring with Kubernetes pod dashboards and Gmail SMTP alerting for 15+ infrastructure components; resolved MySQL binary log disk exhaustion.
- Resolved LXD container internet access and VS Code Remote tunnel connectivity for 5 engineers via **MASQUERADE NAT rules** on bond0 interface.

K3s | HashiCorp Vault | LXD | Ubuntu 22.04 | iptables | fail2ban | cert-manager | CoreDNS | nginx Ingress | Zabbix | GitHub Actions | SonarQube | Trivy | Gitleaks | rsyslog | NVIDIA V100

Backend Engineer

Jan 2024 – Dec 2025

BoolMind · AI & Automation Platform · Remote

- Architected and delivered a production microservices backend for an AI agent platform, handling **10k+ daily API requests** with **sub-100ms P95 latency** using FastAPI and asynchronous Python (asyncio).
- Designed and implemented **gRPC-based inter-service communication** across 6 independent services, reducing internal API overhead by **40%** compared to REST.
- Built Redis-backed job queuing and caching layer, cutting database read load by **60%** and enabling real-time agent state synchronisation.
- Deployed LLM inference infrastructure on 4× Tesla V100 GPU server: vLLM serving Qwen2.5-Coder-32B, FastAPI load-balancing router across 4 GPU instances, nginx API gateway with API key authentication.
- Implemented OWASP Top 10 mitigations, ECDH key exchange, and Attribute-Based Encryption (ABE) for tenant data isolation across the multi-tenant API layer.
- Engineered Stripe payment security hardening after a carding attack: Radar fraud rules, Redis-based rate limiting per IP/card, HMAC inter-service authentication — **zero fraudulent transactions post-deployment**.
- Delivered a Python drop-in client library for the LLM API consumed by 5 frontend developers, reducing integration time from days to hours.

FastAPI | asyncio | gRPC | Redis | PostgreSQL | Python | vLLM | nginx | OWASP Top 10 | ECDH | ABE | HMAC | Stripe Radar | Docker

Junior Backend Engineer

Sep 2022 – Dec 2023

Techleadz · Software Development Agency · Lahore, Pakistan

- Developed and scaled RESTful APIs using Django REST Framework and Node.js (NestJS) for SaaS products with **50k+ registered users**.
- Designed PostgreSQL schemas for multi-tenant SaaS applications, implementing row-level security and optimised indexing strategies reducing query time by **35%**.
- Built MongoDB document models using Beanie ODM for high-write analytics workloads, handling **500k+ daily events**.

- Integrated third-party payment, notification, and authentication services (Stripe, Firebase, OAuth2), delivering end-to-end feature ownership.
- Participated in sprint-based Agile delivery, consistently hitting sprint commitments and contributing to architectural design reviews.
- Deployed and maintained applications on Linux VPS: nginx reverse proxy, SSL, systemd service management.
- Wrote Dockerfiles and Docker Compose configurations for development and staging environments.

Django | Django REST Framework | NestJS | Node.js | PostgreSQL | MongoDB | Beanie | Redis | Stripe | Firebase | OAuth2 | Docker | nginx | systemd

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, Go, Bash
Frameworks	FastAPI, Django, Django REST Framework, Flask, NestJS, Node.js, Express.js
Databases & ORMs	PostgreSQL, MongoDB, Redis, MySQL · SQLAlchemy, Django ORM, Beanie, Mongoose
APIs & Protocols	REST, gRPC, WebSockets, GraphQL, OAuth2, JWT, OpenAPI/Swagger
Cloud & DevOps	AWS (EC2, S3, Lambda, RDS, IAM), GCP, Docker, Kubernetes (K3s), GitHub Actions CI/CD
Container & Orchestration	K3s, LXD, Docker Compose, Helm, nginx Ingress, CoreDNS, cert-manager
Security	OWASP Top 10, ECDH, ABE encryption, JWT hardening, HMAC, SAST/DAST
Secrets & Compliance	HashiCorp Vault (AppRole, dynamic secrets, policies), GDPR data residency
Linux & Networking	Ubuntu 22.04, iptables, fail2ban, MASQUERADE NAT, bond interfaces, SSH hardening, systemd
Monitoring	Zabbix, Grafana, Prometheus, Trivy, Gitleaks, SonarQube, rsyslog
AI / MLOps	vLLM, LLM deployment, FastAPI inference routing, GPU cluster management (NVIDIA V100)

KEY PROJECTS

LLM Inference Platform (Codsy)

Designed and deployed a production-grade multi-GPU LLM inference cluster on 4× Tesla V100 hardware. Built FastAPI load-balancing router, nginx reverse proxy with API key authentication, vLLM engine management, and drop-in Python SDK. Serves an active developer team of 5+.

Zero-Trust Secrets Management (HashiCorp Vault)

Designed and deployed Vault in an LXD container with AppRole auth, policy-based secret access, and dynamic credential rotation for SSH and SonarQube CI/CD pipelines. GDPR data residency compliant.

AI Agent Microservices Platform

Production microservices backend for AI automation — 6 gRPC-connected services, Redis job queuing, ECDH + ABE multi-tenant isolation, sub-100ms P95 latency at 10k+ daily requests.

EDUCATION

Bachelor of Science in Computer Science

2018 – 2022

FAST-NUCES (National University of Computer & Emerging Sciences) · Lahore, Pakistan

Relevant coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems, Software Engineering, Distributed Systems, Cybersecurity

LANGUAGES

English	C1 – Professional Working Proficiency (full technical and business fluency)
Urdu	Native

ADDITIONAL

- **Work Model:** Open to on-site, hybrid, or fully remote positions in Lahore or Pakistan-wide
- **Availability:** Open to freelance projects and full-time roles